RAJALAKSHMI ENGINEERING COLLEGE

RAJALAKSHMI NAGAR, THANDALAM – 602 105



CS23331 - DESIGN AND ANALYSIS OF ALGORITHMS

Laboratory Record Note Book

Name : Nithil Matthew Jude Tamilselvan

Year / Branch / Section : 2nd year, Computer Science and Design

University Register No. : 2116251701036

College Roll No. : 251701036

Semester : 3rd Semester

Academic Year : 2026-2027

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 9:40 PM
State	Finished
Completed on	Saturday, 10 October 2026, 10:44 PM
Time taken	1 hour 4 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Quiz navigation

1

Finish review

Question 1

Correct Mark 1.00 out of 1.00 Flag question

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 0;
    }

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }


    int slow = arr[0];
    int fast = arr[0];

    do {
        slow = arr[slow];
        fast = arr[arr[fast]];
    } while (slow != fast);


    slow = arr[0];
    while (slow != fast) {
        slow = arr[slow];
        fast = arr[fast];
    }

    printf("%d\n", slow);

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 10:46 PM
State	Finished
Completed on	Saturday, 10 October 2026, 10:46 PM
Time taken	38 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Quiz navigation

1

Finish review

Question 1

Correct Mark 1.00 out of 1.00 Flag question

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 0;
    }

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // Phase 1: Finding the intersection point in the cycle
    int slow = arr[0];
    int fast = arr[0];

    do {
        slow = arr[slow];
        fast = arr[arr[fast]];
    } while (slow != fast);

    // Phase 2: Finding the entrance to the cycle (the duplicate number)
    slow = arr[0];
    while (slow != fast) {
        slow = arr[slow];
        fast = arr[fast];
    }

    printf("%d\n", slow);

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 10:48 PM
State	Finished
Completed on	Saturday, 10 October 2026, 10:48 PM
Time taken	34 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void solve() {
    int n1;
    if (scanf("%d", &n1) != 1) return;

    int a[n1];
    for (int i = 0; i < n1; i++) {
        scanf("%d", &a[i]);
    }

    int n2;
    if (scanf("%d", &n2) != 1) return;

    int b[n2];
    for (int j = 0; j < n2; j++) {
        scanf("%d", &b[j]);
    }

    int i = 0;
    int j = 0;
    int first = 1;

    // Two-pointer traversal
    while (i < n1 && j < n2) {
        if (a[i] == b[j]) {
            if (!first) {
                printf(" ");
            }
            printf("%d", a[i]);
            first = 0;
            i++;
            j++;
        } else if (a[i] < b[j]) {
            i++;
        } else {
            j++;
        }
    }
    printf("\n");
}

int main() {
    int t;
    if (scanf("%d", &t) != 1) {
        return 0;
    }

    while (t--) {
        solve();
    }

```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Quiz navigation

1

Finish review

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 10:50 PM
State	Finished
Completed on	Saturday, 10 October 2026, 10:51 PM
Time taken	34 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct Mark 1.00 out of 1.00 Flag question

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void solve() {
    int n1;
    if (scanf("%d", &n1) != 1) return;

    int a[n1];
    for (int i = 0; i < n1; i++) {
        scanf("%d", &a[i]);
    }

    int n2;
    if (scanf("%d", &n2) != 1) return;

    int b[n2];
    for (int j = 0; j < n2; j++) {
        scanf("%d", &b[j]);
    }

    int i = 0;
    int j = 0;
    int first = 1;

    while (i < n1 && j < n2) {
        if (a[i] == b[j]) {
            if (!first) {
                printf(" ");
            }
            printf("%d", a[i]);
            first = 0;
            i++;
            j++;
        } else if (a[i] < b[j]) {
            i++;
        } else {
            j++;
        }
    }
    printf("\n");
}

int main() {
    int t;
    if (scanf("%d", &t) != 1) {
        return 0;
    }

    while (t--) {
        solve();
    }

    return 0;
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Quiz navigation

1

Finish review

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 10:52 PM
State	Finished
Completed on	Saturday, 10 October 2026, 10:53 PM
Time taken	42 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) {
        printf("0\n");
        return 0;
    }

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int k;
    if (scanf("%d", &k) != 1) {
        printf("0\n");
        return 0;
    }

    // Two pointers: i and j both start from the beginning
    int i = 0;
    int j = 1;
    int found = 0;

    while (i < n && j < n) {
        // Enforce the condition i != j
        if (i == j) {
            j++;
            continue;
        }

        int diff = arr[j] - arr[i];

        if (diff == k) {
            found = 1;
            break;
        } else if (diff < k) {
            j++; // Increase difference by moving j to the right
        } else {
            i++; // Decrease difference by moving i to the right
        }
    }

    printf("%d\n", found);

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Quiz navigation

1

Finish review

CS23331-Design and Analysis of Algorithms-2025 Batch-CSD

Started on	Saturday, 10 October 2026, 10:54 PM
State	Finished
Completed on	Saturday, 10 October 2026, 10:55 PM
Time taken	1 min 27 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Quiz navigation

1

Finish review

Question 1

Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1 || n <= 0) {
        printf("0\n");
        return 0;
    }

    int arr[n];
    for (int idx = 0; idx < n; idx++) {
        scanf("%d", &arr[idx]);
    }

    int k;
    if (scanf("%d", &k) != 1) {
        printf("0\n");
        return 0;
    }

    int i = 0;
    int j = 1;
    int found = 0;

    while (i < n && j < n) {
        if (i == j) {
            j++;
            continue;
        }

        int diff = arr[j] - arr[i];

        if (diff == k) {
            found = 1;
            break;
        } else if (diff < k) {
            j++;
        } else {
            i++;
        }
    }

    printf("%d\n", found);

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review